Status	Finished
Started	Wednesday, 21 October 2026, 12:16 PM
Completed	Wednesday, 21 October 2026, 12:20 PM
Duration	4 mins 40 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. Selection
- c. Duplicates
- d. All of the above

The correct answer is: All of the above

Question **2**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 80,150,200,470,40,90,300,400,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **3**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Inserting data into a list
- b. Arranging data in a particular format
- c. Searching for data in a list
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **4**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Rearranging
- b. Searching
- c. Sorting
- d. Merging

The correct answer is: Sorting

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Sorting the entire list sequentially
- b. Dividing the input into smaller subproblems
- c. Combining sorted sublists
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **6**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Heap sort
- b. Insertion sort
- c. Bubble sort
- d. Quick sort

The correct answer is:
Insertion sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It simplifies the sorting process by using only one pass
- c. It avoids the need for recursion
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **8**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Dividing the list into halves
- b. Bubbling up the largest element to its correct position
- c. Sorting the entire list in one pass
- d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **9**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By using the bubble-up method
- b. By comparing elements sequentially
- c. By sorting data in a single pass
- d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **10**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort does not guarantee sorted order
- c. Bubble Sort is difficult to implement
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are combined to form a single sorted list
- c. They are discarded
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **12**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It slows down the process
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **13**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Selection
c. Insertion
d. Extraction

The correct answer is:
Insertion

Question **14**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of duplicate keys
b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
c. A sorting algorithm is stable if it preserves the order of all keys
d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **15**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method

b. Backtracking

c. Divide and conquer

d. Dynamic programming

The correct answer is:
Divide and conquer